Tweet Sentiment Analysis

Classifying Customer Opinions on Apple and Google Products

Phase 4 Project | Natural Language Processing | Angela Mukami, Tafford Pessah, Catherine Nanjala, Mophat Mwangi

9,093	3	3	0.6078
Tweets Analyzed	Sentiment Classes	Models Compared	Best Weighted F1

M-KOPA General

The Business Problem

Technology companies like Apple and Google receive thousands of tweets every day. Their social media and customer experience teams need a fast, reliable way to sort incoming tweets by sentiment - so they can act on what matters most.

Respond Fast

Identify and escalate negative tweets before they become a public issue

Track Positivity

Monitor positive sentiment trends around product launches and campaigns

Filter Noise

Automatically set aside neutral tweets that require no action

Our Solution: Build a model that automatically classifies any tweet as Positive, Negative, or Neutral based on its text content.

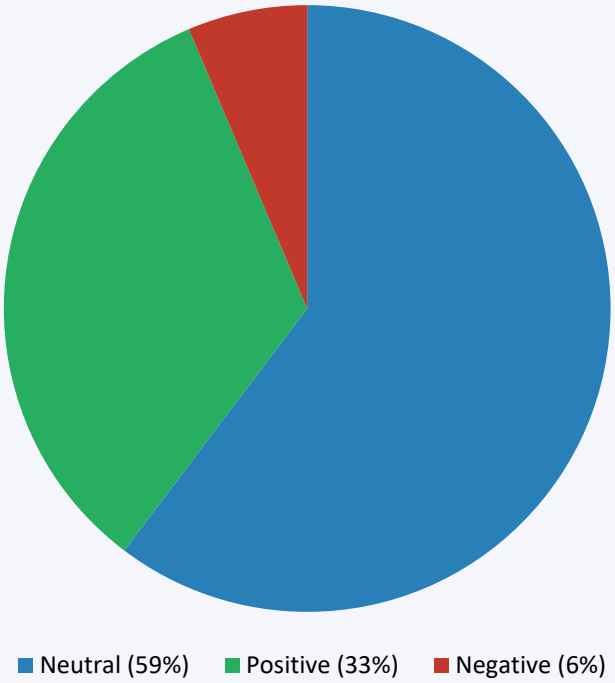
About the Data

9,093
tweets rated by human judges

5,389
neutral tweets (59%)

2,978
positive tweets (33%)

570
negative tweets (6%)



Key challenge: Negative tweets are only 6% of the data. This imbalance directly shaped our modeling strategy.

How We Prepared the Data

1	**Removed noise**	Dropped 1 null tweet and 156 ambiguous "I can't tell" entries that could not be reliably learned
2	**Cleaned the text**	Removed URLs, @mentions, punctuation and numbers. Converted all text to lowercase for consistency
3	**Removed stop words**	Filtered out common words like "the", "is", "and" that carry no sentiment signal and add noise
4	**Converted text to numbers**	Used TF-IDF to score each word by how important it is in a tweet vs across all tweets. Included two-word phrases to capture context like "line ipad" or "app store"
5	**Balanced the classes**	Applied class weighting in models so the rare negative class (6%) receives more attention during training

The Three Models We Tested

We tested three different approaches, each building on the last. All models read the cleaned, TF-IDF weighted tweet text.

Baseline

Naive Bayes

A simple probability-based model. Fast and easy to interpret but treats each word independently and cannot compensate for class imbalance.

Accuracy: 0.6468 | Weighted F1: 0.5983

High accuracy but nearly ignored negative tweets entirely - only 3 out of 171 correctly identified.

Challenger

Logistic Regression

A stronger linear model that can apply higher penalties for misclassifying rare classes. Used class weighting to give more attention to negative tweets.

Accuracy: 0.5401 | Weighted F1: 0.5704

Best negative class recall (93 correct) but sacrificed too much neutral performance.

Winner

Linear SVC

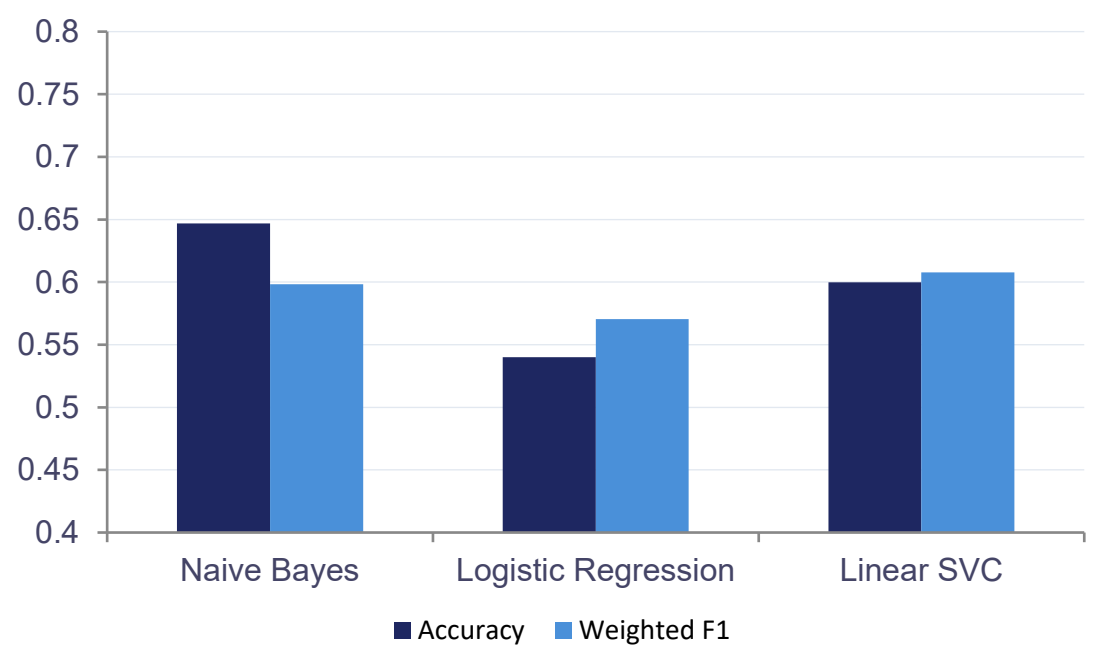
A support vector classifier well suited to high-dimensional text data. Finds the best boundary between classes. Used class weighting throughout.

Accuracy: 0.5998 | Weighted F1: 0.6078

Best overall balance across all three classes. Selected as the final model.

Model Results Comparison

We use Weighted F1 as our primary metric - it accounts for class imbalance and penalises models that ignore minority classes.



Key Insight

- Linear SVC wins on Weighted F1 (0.6078) - our most important metric
- Naive Bayes had the highest accuracy but nearly missed all negative tweets
- Lower accuracy can mean better fairness across classes - not worse performance
- Weighted F1 tells the full story when classes are unequal

What Words Drive Each Sentiment?

We examined which words most strongly predict each sentiment class. This helps validate that the model is learning real signals, not noise.

Negative Tweets

- doesnt
- battery
- line ipad
- app store
- long
- think
- iphone
- apples

Words signal frustration with product performance, wait times, and app access issues.

Neutral Tweets

- link
- designing
- codes
- making
- music
- location
- mobile
- free

Purely informational words. Tweets sharing news or product details with no emotion.

Positive Tweets

- cool
- great
- awesome
- fun
- smart
- wow
- wins
- googles

Clear enthusiasm markers. Conference energy from SXSW also visible in this class.

Limitations to Be Aware Of

Very few negative examples

Only 6% of tweets are negative. Even with class weighting, the model has limited data to learn from for this critical class. More labeled negative examples would directly improve performance.

Tweets are short and noisy

Tweets use slang, abbreviations, and informal language that standard cleaning does not fully handle. Some preprocessing artifacts such as merged words appeared in the top features.

Context and sarcasm are missed

The model reads tweets as a bag of words. It cannot detect sarcasm, irony, or context-dependent meaning. A tweet like 'great, another crash' would likely be misclassified as positive.

Recommendations

1	**Deploy the model for first-pass triage**	With a Weighted F1 of 0.6078, the Linear SVC model provides a meaningful filter on incoming tweets. It will significantly reduce the volume requiring manual review, even if not perfect.
2	**Collect more negative tweet examples**	Negative tweets are the most business-critical class and the most underrepresented. Targeted labeling of negative tweets would have the highest return on investment for model improvement.
3	**Retrain the model periodically**	Sentiment language evolves with product cycles and current events. The model should be updated with fresh labeled data at regular intervals to remain accurate and relevant.
4	**Consider a two-stage approach**	A practical first deployment could classify tweets as 'needs attention' (negative) vs 'does not need attention' (positive or neutral). This binary split is simpler and more reliable with the current data.

Summary

- Analyzed 9,093 tweets about Apple and Google products to classify sentiment as positive, negative, or neutral
- Built and compared three models: Naive Bayes, Logistic Regression, and Linear SVC
- Linear SVC selected as the final model with the best Weighted F1 score of 0.6078
- Model explainability confirmed the classifier is learning real sentiment signals
- Key next step: collect more negative tweet examples to improve the most critical class